

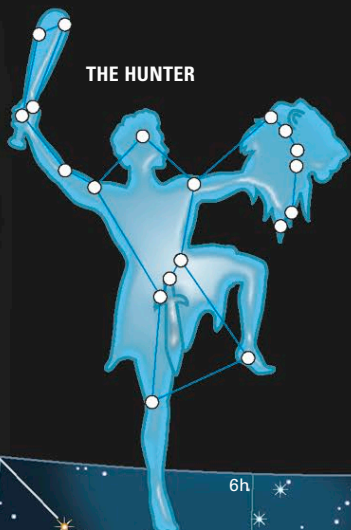
The real treasures of this constellation lie in the area around Orion's sword. NGC 1981, for example, appears as a large, scattered cluster of stars through binoculars; its brightest stars are of 6th magnitude. NGC 1977 is an elongated patch of nebulosity surrounding the stars 42 and 45 Orionis. Nearby is the Orion Nebula, an enormous star-forming cloud of gas, 1,500 light-years away, covering an area of sky wider than two diameters of a full moon. Its glowing gas appears multicolored on photographs and CCD images, yet visually it looks only gray-green because the eye is not sensitive to colors in

faint objects. On clear nights, it appears to the naked eye as a hazy patch of light and is obvious through any form of optical aid. An extension of the Orion Nebula bears a separate number, M43, but both are part of the same cloud. At the center of M42 lies a multiple star, Theta-1 (θ^1) Orionis (see p.281), better known as the Trapezium because it appears as a group of four stars of 5th to 8th magnitude when seen through a small telescope. To one side of the nebula lies Theta-2 (θ^2) Orionis, a double star with components of 5th and 6th magnitudes that can be separated with binoculars. At the tip of Orion's sword lies Iota (ι) Orionis, a double, with components of 3rd and 7th magnitudes, divisible with a small telescope. Struve 747 is a wider double star nearby, with components of 5th and 6th magnitudes.

Even more impressive is the multiple star Sigma (σ) Orionis (p.281). A small telescope shows that the main 4th-magnitude star has two 7th-magnitude companions on one side and a closer 9th-magnitude component on the other.

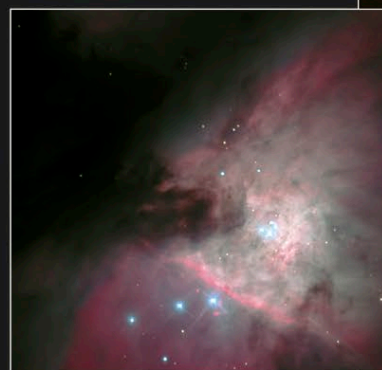
Extending from the belt star Zeta (ζ) Orionis is a strip of bright nebulosity, IC 434, against which is silhouetted the Horsehead Nebula (see p.240). This is probably the best-known dark nebula in the sky, and it shows well on photographs. To see it visually requires a large telescope and a dark viewing site.

THE HUNTER



THE HORSEHEAD NEBULA 𐂔 𐂔

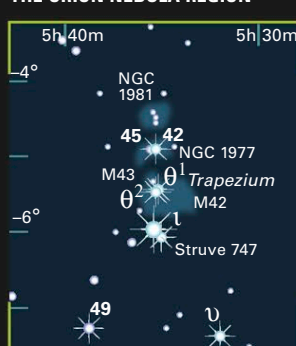
Looking like a knight in a celestial chess game, the Horsehead Nebula is a curiously shaped dark dust cloud silhouetted against IC 434, a backdrop of glowing hydrogen. It lies to the south of Zeta (ζ) Orionis (center left) in Orion's belt.



THE TRAPEZIUM 𐂔 𐂔

At the heart of the Orion Nebula lies a multiple star called the Trapezium (θ^1) (center, right). Its four stars are visible through small telescopes, but with larger apertures, two additional stars can also be seen.

THE ORION NEBULA REGION



MAGNITUDE KEY



THE ORION NEBULA 𐂔 𐂔

To the naked eye and through binoculars, the Orion Nebula (M42) appears only as a misty patch south of Orion's belt, its heart lit up by newborn stars. Its full beauty and its pinkish color become apparent only on photographs and CCD images such as this.